

Algebra over a field K

Groups, rings, fields, vector spaces, modules, and bilinear maps

An algebra combines a vector-space structure with a compatible multiplication. We build toward this definition from binary operations, through groups, rings, fields, and vector spaces. Modules extend scalar multiplication from fields to rings. Linear and bilinear maps then provide the language needed to formulate the compatibility precisely. Important geometric examples are the algebras of smooth functions and smooth function germs.

The labels attached to axioms identify the structure under discussion. Thus $(G1)$ denotes a group axiom, $(R1)$ denotes a ring axiom, and $(V1)$ denotes a vector-space axiom. These labels are local conventions rather than universal names, but they make dependencies between definitions visible. We write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

1 Binary operations

Definition 1.1: Binary operation

Let S be a set. A *binary operation on S* is a function

$$* : S \times S \longrightarrow S, \quad (a, b) \longmapsto a * b.$$

The codomain S means that S is closed under the operation $*$. In other words, if $a, b \in S$, then $a * b \in S$.

The symbol $*$ carries no fixed meaning. It can denote addition, multiplication, composition, or another operation. What matters is the map and the properties it satisfies.

Definition 1.2: Standard properties of a binary operation

Let $*$ be a binary operation on S . The following properties will be used repeatedly.

(O1) **Associativity.** For every $a, b, c \in S$,

$$(a * b) * c = a * (b * c).$$

(O2) **Identity.** There exists $e \in S$ such that, for every $a \in S$,

$$e * a = a * e = a.$$

(O3) **Existence of inverses.** Relative to an identity e , every $a \in S$ has a two-sided inverse, that is, an element $b \in S$ such that

$$a * b = b * a = e.$$

(O4) **Commutativity.** For every $a, b \in S$,

$$a * b = b * a.$$

An identity, when it exists, is unique. If the operation is also associative, inverses are unique whenever they exist. In this setting the notation a^{-1} is unambiguous. These assertions are proved in Proposition 2.4.

Distributivity relates two operations. If $+$ and $\cdot$ are binary operations on the same set, then multiplication distributes over addition on the left and on the right, respectively, when

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc$$

for all a, b, c . Both laws will be required in the definition of a ring.

Closure is built into the definition

One sometimes sees closure listed as a separate axiom. In the notation $* : S \times S \rightarrow S$, closure is already encoded by the codomain. A rule such as subtraction on $\mathbb{N}$ is not a binary operation on $\mathbb{N}$ because the result can lie outside $\mathbb{N}$.

Example 1.3: Composition of maps

Let X be a set and let S be the set of all functions $X \rightarrow X$. Composition defines a binary operation on S by

$$(f, g) \mapsto f \circ g.$$

It is associative. It has the identity map id_X as an identity element. It is generally not commutative.

Key idea. A binary operation is a function with two inputs from the same set and one output in that set. Its additional properties are separate statements that must be proved or assumed.

2 Semigroups, monoids, and groups

The next structures are obtained by imposing progressively stronger axioms on one binary operation.

Definition 2.1: Semigroup

A *semigroup* is a nonempty set S with a binary operation $*$ satisfying the following axiom.

(S1) **Associativity.** For every $a, b, c \in S$,

$$(a * b) * c = a * (b * c).$$

Definition 2.2: Monoid

A *monoid* is a set M with a binary operation $*$ satisfying the following axioms.

(M1) **Associativity.** For every $a, b, c \in M$,

$$(a * b) * c = a * (b * c).$$

(M2) **Identity.** There exists $e \in M$ such that, for every $a \in M$,

$$e * a = a * e = a.$$

Definition 2.3: Group and abelian group

A *group* is a set G with a binary operation $*$ satisfying the following axioms.

(G1) **Associativity.** For every $a, b, c \in G$,

$$(a * b) * c = a * (b * c).$$

(G2) **Identity.** There exists $e \in G$ such that, for every $a \in G$,

$$e * a = a * e = a.$$

(G3) **Inverses.** For every $a \in G$, there exists $a^{-1} \in G$ such that

$$a * a^{-1} = a^{-1} * a = e.$$

The group is *abelian* if it also satisfies

(G4) **Commutativity.** For every $a, b \in G$,

$$a * b = b * a.$$

The definitions give the implication chain

$$\text{abelian group} \implies \text{group} \implies \text{monoid} \implies \text{semigroup}.$$

Each arrow means that the structure on the left satisfies every axiom required by the structure on the right. The reverse implications fail in general.

Proposition 2.4: Uniqueness of the identity and inverses

A binary operation has at most one two-sided identity. If the operation is associative and has an identity, each element has at most one two-sided inverse. In particular, a group's identity and inverses are unique.

Proof. Suppose e and e' are both identities. Then

$$e = e * e' = e'.$$

If the operation is associative and b and c are both inverses of a , then

$$b = b * e = b * (a * c) = (b * a) * c = e * c = c.$$

Thus identities are unique, and associativity ensures uniqueness of inverses whenever they exist. $\square$

Example 2.5: Additive examples

The pair $(\mathbb{N}_0, +)$ is a commutative monoid but not a group. A positive integer has no additive inverse in $\mathbb{N}_0$. The pair $(\mathbb{Z}, +)$ is an abelian group. Its identity is 0, and the inverse of $n \in \mathbb{Z}$ is $-n$.

3 Rings

A ring places two binary operations on one set. Their usual names are addition and multiplication. The notation alone does not imply any axioms.

Definition 3.1: Ring

A *ring* is a set A with two binary operations

$$+ : A \times A \longrightarrow A, \quad \cdot : A \times A \longrightarrow A,$$

satisfying the following axioms.

(R1) **Additive abelian group.** The additive structure $(A, +)$ is an abelian group.

(R2) **Associative multiplication.** For every $a, b, c \in A$,

$$(ab)c = a(bc).$$

(R3) **Distributivity.** For every $a, b, c \in A$,

$$(a + b)c = ac + bc, \quad a(b + c) = ab + ac.$$

Axiom (R1) includes all four group axioms for addition. Thus addition is associative and commutative, A has an additive identity 0_A , and every $a \in A$ has an additive inverse $-a$. We write ab for $a \cdot b$.

Since A is nonempty by (R1), axiom (R2) makes $(A, \cdot)$ a semigroup. Axiom (R3) relates multiplication to addition. Without commutativity of multiplication, neither distributive law follows from the other, so both are stated.

Definition 3.2: Additional ring properties

A ring may satisfy either of the following additional axioms.

(R4) **Multiplicative identity.** There exists $1_A \in A$ such that, for every $a \in A$,

$$1_A a = a 1_A = a.$$

A ring satisfying (R4) is called *unital*.

(R5) **Commutative multiplication.** For every $a, b \in A$,

$$ab = ba.$$

A ring satisfying (R5) is called *commutative*.

If (R4) holds, then $(A, \cdot)$ is a monoid. The zero element and the unit must not be confused. The *zero ring* consists of the single element 0_A . Its multiplicative identity is also 0_A . In every other unital ring, $0_A \neq 1_A$. Indeed, $1_A = 0_A$ would imply $a = 1_A a = 0_A a = 0_A$ for every a , by the following proposition.

Proposition 3.3: Elementary consequences of the ring axioms

For every $a, b \in A$,

$$0_A a = a 0_A = 0_A, \quad (-a)b = -(ab), \quad a(-b) = -(ab).$$

Proof. Using distributivity,

$$0_A a = (0_A + 0_A)a = 0_A a + 0_A a.$$

Adding the additive inverse of $0_A a$ to both sides gives $0_A a = 0_A$. The proof of $a 0_A = 0_A$ is analogous. Also,

$$ab + (-a)b = (a + (-a))b = 0_A b = 0_A,$$

so $(-a)b$ is the additive inverse of ab . The last identity is analogous. $\square$

Example 3.4: Integers and matrices

The integers $\mathbb{Z}$ form a commutative unital ring.

For a field K and $n \geq 1$, let $M_n(K)$ denote the set of $n \times n$ matrices with entries in K . It is a unital ring under matrix addition and matrix multiplication, with unit I_n . For $n \geq 2$, it is noncommutative. For example, in $M_2(K)$,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad PQ = Q, \quad QP = 0.$$

The same example embeds into larger matrices by filling the other entries with zeros.

Convention concerning units. Here a ring need not have a multiplicative identity. The adjective *unital* states this additional assumption. Another common convention includes a multiplicative identity in the definition of a ring, so the chosen convention should always be kept in mind.

4 Fields

A field is a ring in which the nonzero elements also form an abelian group under multiplication.

Definition 4.1: Field

A *field* is a set K with binary operations $+, \cdot : K \times K \rightarrow K$ satisfying the following axioms.

- (F1) **Additive abelian group.** The additive structure $(K, +)$ is an abelian group.
- (F2) **Multiplicative abelian group.** The multiplicative structure $(K \setminus \{0_K\}, \cdot)$ is an abelian group.
- (F3) **Distributivity.** For every $r, s, t \in K$,

$$r(s + t) = rs + rt, \quad (r + s)t = rt + st.$$

Axiom (F2) includes multiplicative associativity, commutativity, the identity 1_K , and an inverse r^{-1} for every nonzero $r \in K$. It also implies $0_K \neq 1_K$ because the identity 1_K belongs to $K \setminus \{0_K\}$. Closure of this group asserts that a product of nonzero elements is nonzero.

Proposition 4.2: Equivalent ring formulation of a field

A field is equivalently a commutative unital ring K with $1_K \neq 0_K$ in which every nonzero element has a multiplicative inverse.

Proof. Assume (F1)–(F3). Additive cancellation and distributivity give $0_K r = r 0_K = 0_K$ by the same argument as in Proposition 3.3. That argument does not use multiplicative associativity. By (F2), multiplication is associative and commutative on the nonzero elements and has identity 1_K . The identities $0_K r = r 0_K = 0_K$ then show that associativity, commutativity, and the identity law also hold whenever one of the factors is 0_K . Hence these properties hold on all of K . Thus K is a commutative unital ring with $1_K \neq 0_K$, and (F2) supplies the inverses.

Conversely, let K be such a ring. If $r, s \neq 0_K$ and $rs = 0_K$, then

$$s = (r^{-1}r)s = r^{-1}(rs) = 0_K,$$

a contradiction. Therefore the nonzero elements are closed under multiplication. An inverse of a nonzero element is itself nonzero, since $r 0_K = 0_K \neq 1_K$. The ring laws and the assumed inverses now make $K \setminus \{0_K\}$ an abelian group. The other field axioms are ring axioms. $\square$

The condition $1_K \neq 0_K$ is essential. In the zero ring there are no nonzero elements, so the requirement that every nonzero element have an inverse would hold vacuously.

Example 4.3: Fields and nonfields

The sets

$$\mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}$$

are fields with their usual operations.

The integers $\mathbb{Z}$ are not a field because $2^{-1} = 1/2$ does not belong to $\mathbb{Z}$.

For every prime number p , the residue classes $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ form a finite field.

Where the group structures occur

Structure	Additive part	Multiplicative part
Ring A	$(A, +)$ is an abelian group	$(A, \cdot)$ is a semigroup
Unital ring A	$(A, +)$ is an abelian group	$(A, \cdot)$ is a monoid
Field K	$(K, +)$ is an abelian group	$(K \setminus \{0\}, \cdot)$ is an abelian group

Key idea. The field K used in linear algebra is itself an algebraic structure. Its elements become the scalars acting on a vector space.

5 Vector spaces over a field

Fix a field K . A vector-space structure consists of an addition on a set V and an action of the scalars in K on the elements of V .

Definition 5.1: K -vector space

A *vector space over K* is a set V equipped with vector addition

$$+ : V \times V \longrightarrow V, \quad (u, v) \longmapsto u + v,$$

and scalar multiplication

$$K \times V \longrightarrow V, \quad (r, v) \longmapsto rv,$$

satisfying the following axioms for all $r, s \in K$ and $u, v \in V$.

(V1) **Additive abelian group.** The additive structure $(V, +)$ is an abelian group.

(V2) **Distributivity over vector addition.**

$$r(u + v) = ru + rv.$$

(V3) **Distributivity over scalar addition.**

$$(r + s)v = rv + sv.$$

(V4) **Compatibility with multiplication in K .**

$$(rs)v = r(sv).$$

(V5) **Scalar identity.**

$$1_K v = v.$$

The phrase *over K* identifies the field of scalars. It does not mean that V is a subset of K .

The two operations of a vector space

Vector addition	$V \times V \rightarrow V$	$(u, v) \mapsto u + v$
Scalar multiplication	$K \times V \rightarrow V$	$(r, v) \mapsto rv$

Scalar multiplication is called an *external operation*. It takes a scalar from K and a vector from V . In general, it is not a binary operation on V . The roles of K and V remain distinct even when the underlying sets coincide, as in the vector space K over itself. Similarly, $+$ denotes both scalar addition and vector addition. Its meaning is determined by the inputs.

Example 5.2: Standard vector spaces

The Cartesian space K^n is a vector space over K with componentwise addition and scalar multiplication. The polynomial space $K[x]$ is a vector space over K with the usual coefficientwise operations. Its vectors are polynomials of arbitrary finite degree. Constant polynomials identify a copy of K inside this space. The matrix space $M_n(\mathbb{C})$ is a vector space over both $\mathbb{C}$ and $\mathbb{R}$. Its dimensions are

$$\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2, \quad \dim_{\mathbb{R}} M_n(\mathbb{C}) = 2n^2.$$

Proposition 5.3: Elementary consequences of the vector-space axioms

For every $r \in K$ and $v \in V$,

$$0_K v = 0_V, \quad r 0_V = 0_V, \quad (-1_K)v = -v.$$

Proof. The first identity follows from

$$0_K v = (0_K + 0_K)v = 0_K v + 0_K v.$$

The second follows similarly from $r(0_V + 0_V) = r 0_V + r 0_V$. Finally,

$$v + (-1_K)v = 1_K v + (-1_K)v = (1_K - 1_K)v = 0_K v = 0_V.$$

Hence $(-1_K)v$ is the additive inverse of v . □

6 Modules over a ring

A vector space has scalars in a field. The same pattern of addition and scalar multiplication makes sense when the scalars form a ring. We require the ring to be commutative and unital, as in the definition below.

Definition 6.1: R -module

Let R be a commutative unital ring. An R -module is a set M with an addition $M \times M \rightarrow M$ and a scalar multiplication

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

satisfying the following axioms for all $r, s \in R$ and $m, n \in M$.

(Mod1) **Additive abelian group.** $(M, +)$ is an abelian group.

(Mod2) **Distributivity over module addition.** $r(m + n) = rm + rn$.

(Mod3) **Distributivity over scalar addition.** $(r + s)m = rm + sm$.

(Mod4) **Compatibility with ring multiplication.** $(rs)m = r(sm)$.

(Mod5) **Scalar identity.** $1_R m = m$.

The elements of M are sometimes called vectors, even though R may not be a field. If $R = K$ is a field, the module axioms are exactly the vector-space axioms (V1)–(V5). Every K -vector space is therefore a K -module, and conversely. The change is in the scalars: a nonzero element of a general ring need not have an inverse, so one cannot divide a module element by it.

The same cancellation arguments as in Proposition 5.3 show that $0_R m = 0_M$, $r 0_M = 0_M$, and $(-r)m = -(rm)$. These statements do not require inverses of nonzero scalars.

Example 6.2: Modules over the integers

The ring $\mathbb{Z}$ acts on every abelian group M by repeated addition: $nm = m + \cdots + m$ for $n > 0$, $0m = 0_M$, and $(-n)m = -(nm)$. This action makes M a $\mathbb{Z}$ -module. In particular, $\mathbb{Z}/n\mathbb{Z}$ is a $\mathbb{Z}$ -module for $n \geq 2$. Here $n[1] = [0]$ although $[1] \neq [0]$. A nonzero integer can therefore annihilate a nonzero module element.

Definition 6.3: Module homomorphism

For R -modules M and N , a map $T : M \rightarrow N$ is an R -module homomorphism if, for all $m, n \in M$ and $r \in R$,

$$T(m + n) = T(m) + T(n), \quad T(rm) = rT(m).$$

When R is a field, this is a linear map of vector spaces.

A module is *free* if it has a basis in the familiar sense: every element can be written *uniquely* as a finite R -linear combination of basis elements. For example, R^n is free with its standard basis. Modules need not be free. The $\mathbb{Z}$ -module $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 2$ has no basis: its nonzero elements have finite additive order, whereas each element of a basis of a free $\mathbb{Z}$ -module has infinite additive order.

Key idea. A module is a vector-space-like structure whose scalars belong to a commutative unital ring. It may have no basis, but its scalar multiplication still obeys the familiar distributive and identity laws.

7 Linear and bilinear maps

The multiplication of an algebra must be compatible with the vector-space structure. This compatibility is expressed through linearity and bilinearity.

A vector space need not have an internal product

The vector-space axioms do not define a multiplication $V \times V \rightarrow V$. A quantum state space $\mathcal{H}$ is a complex vector space, but two state vectors do not generally have a product that lies again in $\mathcal{H}$.

Definition 7.1: Linear map

Let V and W be vector spaces over the same field K . A map $T : V \rightarrow W$ is K -linear if it satisfies the following axioms.

(L1) **Additivity.** For every $u, v \in V$,

$$T(u + v) = T(u) + T(v).$$

(L2) **Homogeneity.** For every $r \in K$ and $v \in V$,

$$T(rv) = rT(v).$$

The two conditions can be combined into.

$$T(ru + sv) = rT(u) + sT(v)$$

for all $r, s \in K$ and $u, v \in V$.

Definition 7.2: Bilinear map

Let U , V , and W be vector spaces over K . A map

$$B : U \times V \longrightarrow W$$

is K -bilinear if it satisfies the following conditions for all $r, s \in K$, $u, u_1, u_2 \in U$, and $v, v_1, v_2 \in V$.

(BL1) **Linearity in the first argument.** For every fixed $v \in V$, the map $u \mapsto B(u, v)$ is K -linear. Equivalently,

$$B(ru_1 + su_2, v) = rB(u_1, v) + sB(u_2, v).$$

(BL2) **Linearity in the second argument.** For every fixed $u \in U$, the map $v \mapsto B(u, v)$ is K -linear. Equivalently,

$$B(u, rv_1 + sv_2) = rB(u, v_1) + sB(u, v_2).$$

The word *separately* is important. Bilinearity means linearity in one argument while the other argument is held fixed. It does not mean that B is linear as a map from the product vector space $U \times V$ to W . Indeed, bilinearity gives $B(u, 0) = B(0, v) = 0$. If B were also linear on the product, whose operations are componentwise, then

$$B(u, v) = B((u, 0) + (0, v)) = B(u, 0) + B(0, v) = 0.$$

Thus only the zero bilinear map is also linear on the product vector space.

Example 7.3: Dot product and matrix multiplication

For $n \geq 2$, the Euclidean dot product

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}, \quad (u, v) \longmapsto u \cdot v$$

is bilinear. It is not an internal multiplication on $\mathbb{R}^n$ because its output lies in $\mathbb{R}$ rather than $\mathbb{R}^n$.

Matrix multiplication

$$M_n(K) \times M_n(K) \longrightarrow M_n(K)$$

is bilinear and internal. It is also associative.

The cross product

The cross product $\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is bilinear and internal, but it is not associative. For standard basis vectors e_1, e_2, e_3 ,

$$(e_1 \times e_1) \times e_2 = 0, \quad e_1 \times (e_1 \times e_2) = e_1 \times e_3 = -e_2.$$

It therefore does not define an associative algebra. Instead, the cross product defines a *Lie algebra bracket* on $\mathbb{R}^3$: it is bilinear, antisymmetric, and satisfies the Jacobi identity

$$u \times (v \times w) + v \times (w \times u) + w \times (u \times v) = 0.$$

Complex inner products

The standard complex inner product, with the convention $\langle u, v \rangle = \sum_j \bar{u}_j v_j$, satisfies

$$\langle ru, sv \rangle = \bar{r} s \langle u, v \rangle.$$

It is conjugate-linear in the first argument and linear in the second, hence *sesquilinear*, rather than $\mathbb{C}$ -bilinear. A complex inner product should therefore not be substituted for a bilinear multiplication.

Key idea. For an algebra, the candidate multiplication must first have the correct type $A \times A \rightarrow A$. It must then be bilinear and associative.

8 Algebras over a field

Let A be a vector space over a field K . An algebra structure adds an internal multiplication to A and requires that multiplication to respect the linear structure. Thus an algebra combines the additive and scalar structure of a vector space with an internal product.

In this chapter, *algebra* means *associative algebra* unless stated otherwise. A multiplicative identity is not required, so the word *algebra* alone does not imply unitality. More generally, one may study vector spaces with bilinear products that are not associative, as the cross product illustrates.

Definition 8.1: Associative algebra over K

An *associative algebra over K* is a K -vector space A equipped with a multiplication map

$$\mu : A \times A \longrightarrow A, \quad \mu(a, b) = ab,$$

satisfying the following axioms.

(A1) **Associativity.** For every $a, b, c \in A$,

$$(ab)c = a(bc).$$

(A2) **Distributivity (additivity in both arguments).** For every $a, b, c \in A$,

$$(a + b)c = ac + bc, \quad a(b + c) = ab + ac.$$

(A3) **Homogeneity in both arguments.** For every $r \in K$ and $a, b \in A$,

$$(ra)b = r(ab), \quad a(rb) = r(ab).$$

Key idea. A K -algebra combines a K -vector space with an associative internal K -bilinear multiplication $A \times A \rightarrow A$.

Axioms (A2) and (A3) together say that the multiplication map μ is K -bilinear. Thus the definition can be compressed to one sentence.

Bilinear formulation

An associative K -algebra is a K -vector space equipped with an associative K -bilinear multiplication $A \times A \rightarrow A$.

In particular, homogeneity in the two arguments implies

$$(ra)(sb) = (rs)(ab) \quad (r, s \in K, a, b \in A).$$

This does not impose commutativity of the algebra product. The order of a and b is unchanged.

Definition 8.2: Additional algebra properties

An associative K -algebra may satisfy either of the following additional axioms.

(A4) **Multiplicative identity.** There exists $1_A \in A$ such that $1_A a = a 1_A = a$ for every $a \in A$. The algebra is then called *unital*.

(A5) **Commutativity.** One has $ab = ba$ for all $a, b \in A$. The algebra is then called *commutative*.

Proposition 8.3: Equivalent ring formulation

An associative K -algebra is equivalently a ring A that is also a K -vector space, using the same addition, and whose ring multiplication satisfies

$$r(ab) = (ra)b = a(rb)$$

for every $r \in K$ and $a, b \in A$.

Proof. Suppose first that A is a K -vector space satisfying (A1), (A2), and (A3). Axiom (V1) says that $(A, +)$ is an abelian group. Axiom (A1) gives associativity of multiplication. Axiom (A2) gives both distributive laws. Therefore A is a ring. Axiom (A3) gives the required compatibility with scalars.

Conversely, suppose that A is a ring and a K -vector space with the stated compatibility. Ring distributivity gives (A2). The compatibility condition gives (A3). Ring associativity gives (A1). Hence A is an associative K -algebra. $\square$

How the algebra axioms fit together

The vector-space structure already makes $(A, +)$ an abelian group. Associativity (A1) and distributivity (A2) then give the ring structure. Homogeneity (A3) is the additional condition that ties the ring multiplication to scalar multiplication by K . Thus a K -algebra may be viewed as a ring and a K -vector space whose two structures are compatible.

9 Examples and nonexamples

The three operations on a K -algebra

Addition in A	$A \times A \rightarrow A$	$(a, b) \mapsto a + b$
Multiplication in A	$A \times A \rightarrow A$	$(a, b) \mapsto ab$
Scalar multiplication	$K \times A \rightarrow A$	$(r, a) \mapsto ra$

9.1 Canonical examples

Example 9.1: The field as an algebra over itself

Every field K is a one-dimensional commutative unital algebra over itself. The algebra multiplication is the field multiplication, and

$$\dim_K K = 1.$$

Example 9.2: Polynomial and matrix algebras

The polynomial space $K[x]$ is a commutative unital K -algebra under ordinary polynomial multiplication. The matrix space $M_n(K)$ is a unital K -algebra under matrix multiplication. For $n \geq 2$, it is noncommutative.

Example 9.3: An algebra without an identity

The polynomials with zero constant term form the K -algebra

$$xK[x] = \{xf(x) : f(x) \in K[x]\}.$$

This set is closed under addition, scalar multiplication, and polynomial multiplication. It has no multiplicative identity. Indeed, a nonzero $e(x) \in xK[x]$ has degree at least one, so $e(x)x$ has degree at least two and cannot equal x . The zero polynomial cannot be an identity either.

Example 9.4: Endomorphism algebra

Let V be a vector space over K . The space

$$\text{End}_K(V) = \{T : V \rightarrow V \mid T \text{ is } K\text{-linear}\}$$

is a vector space with pointwise operations

$$(S + T)(v) = S(v) + T(v), \quad (rT)(v) = rT(v).$$

It becomes a unital K -algebra under composition,

$$(S, T) \mapsto S \circ T.$$

Linearity of the operators makes composition bilinear, and composition is associative. Its identity is id_V . If $\dim_K V = n < \infty$, a choice of basis gives a K -algebra isomorphism

$$\text{End}_K(V) \cong M_n(K).$$

This identification depends on the chosen basis.

9.2 Useful nonexamples

- A vector space without a specified internal multiplication is not yet an algebra.
- For $n \geq 2$, the dot product on $\mathbb{R}^n$ is not an algebra multiplication because it takes values in $\mathbb{R}$ rather than $\mathbb{R}^n$. For $n = 1$, under the usual identification $\mathbb{R}^1 = \mathbb{R}$, it is ordinary multiplication and does define an algebra.
- The cross product on $\mathbb{R}^3$ is bilinear and internal, but it is not associative.
- For $n \geq 2$, Hermitian $n \times n$ matrices form a real vector space but are not closed under ordinary matrix multiplication. Here $H^\dagger = \overline{H}^\top$, and H is Hermitian when $H^\dagger = H$. For Hermitian H, J , one has $(HJ)^\dagger = JH$, so HJ is Hermitian exactly when H and J commute. For example,

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad HJ = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The first two matrices are Hermitian, whereas their product is not.

10 Smooth functions, germs, and algebra homomorphisms

The algebraic structures above arise naturally in differential geometry. Smooth functions provide a basic example, and passing to germs records only their local behaviour near a point.

Example 10.1: The algebra of smooth functions

For a nonempty open set $U \subseteq \mathbb{R}^n$, let $C^\infty(U)$ denote the set of smooth real-valued functions on U . Define

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x), \quad (rf)(x) = rf(x),$$

for $x \in U$ and $r \in \mathbb{R}$. These functions are smooth, and the algebra identities follow by evaluating them at each x and using the corresponding identities in $\mathbb{R}$. Thus $C^\infty(U)$ is a commutative unital $\mathbb{R}$ -algebra. Its zero and identity are the constant functions 0 and 1.

Example 10.2: Smooth vector fields form a module

For an open set $U \subseteq \mathbb{R}^n$, a smooth vector field can be written $X(x) = (X^1(x), \dots, X^n(x))$, where each component X^i is smooth. Let $\mathfrak{X}(U)$ denote the set of these fields. If $f \in C^\infty(U)$, define fX pointwise by

$$(fX)(x) = f(x)X(x).$$

The components fX^i are smooth. Pointwise addition of fields and this scalar multiplication satisfy the module axioms, so $\mathfrak{X}(U)$ is a module over the commutative unital ring $C^\infty(U)$. In particular, $(fX)_p = f(p)X_p$: the field fX is obtained by scaling X separately at each point p . The notation fX denotes a vector field, while $(fX)_p$ denotes its value at p .

Definition 10.3: Smooth function germ

Fix $p \in \mathbb{R}^n$. Consider pairs (U, f) , where U is an open neighbourhood of p and $f \in C^\infty(U)$. Set

$$(U, f) \sim (V, g) \iff f|_W = g|_W \text{ on some open neighbourhood } W \subseteq U \cap V \text{ of } p.$$

This is an equivalence relation. Its equivalence classes are the *smooth function germs at p* . The class of (U, f) is denoted by $[f]_p$. The set of these classes is denoted by $C_p^\infty(\mathbb{R}^n)$.

Reflexivity and symmetry follow directly from equality of functions. For transitivity, if $f = g$ on a neighbourhood W_1 and $g = h$ on a neighbourhood W_2 , then $f = h$ on $W_1 \cap W_2$, which is again an open neighbourhood of p . A germ records agreement near p , not merely the value at p . For example, $f(x) = 0$ and $g(x) = x$ on $\mathbb{R}$ have the same value at 0 but distinct germs there.

Proposition 10.4: The algebra of smooth germs

The set $C_p^\infty(\mathbb{R}^n)$ is a commutative unital $\mathbb{R}$ -algebra with

$$[f]_p + [g]_p = [f + g]_p, \quad [f]_p [g]_p = [fg]_p, \quad r[f]_p = [rf]_p.$$

When f and g have different domains, the sum and product are formed on their intersection. The zero and identity are $[0]_p$ and $[1]_p$.

Proof. Suppose $[f]_p = [\tilde{f}]_p$ and $[g]_p = [\tilde{g}]_p$. Choose open neighbourhoods W_f and W_g on which the respective representatives agree. On $W_f \cap W_g$,

$$f + g = \tilde{f} + \tilde{g}, \quad fg = \tilde{f}\tilde{g}, \quad rf = r\tilde{f}.$$

Consequently the resulting germs do not depend on the representatives. Every vector-space and algebra identity now follows from the corresponding identity for smooth functions, after restricting the finitely many representatives to a common neighbourhood. $\square$

Definition 10.5: Algebra homomorphism

Let A and B be K -algebras. A K -algebra homomorphism is a K -linear map $\varphi : A \rightarrow B$ that preserves multiplication,

$$\varphi(ab) = \varphi(a)\varphi(b) \quad (a, b \in A).$$

If A and B are unital, the homomorphism is called *unital* when it also satisfies $\varphi(1_A) = 1_B$.

For unital algebras, preservation of the multiplicative identity is an additional condition in the convention used here. It does not follow from linearity and preservation of products. For example, the zero map $\mathbb{R} \rightarrow \mathbb{R}$ has both properties but does not preserve 1.

Example 10.6: Evaluation at a point

The map

$$\text{ev}_p : C_p^\infty(\mathbb{R}^n) \longrightarrow \mathbb{R}, \quad [f]_p \longmapsto f(p),$$

is well defined because equivalent representatives agree at p . It is $\mathbb{R}$ -linear, satisfies $\text{ev}_p([f]_p[g]_p) = f(p)g(p)$, and sends $[1]_p$ to 1. Hence it is a unital algebra homomorphism.

Connection with derivations at a point

The product in the germ algebra is the product that appears in the Leibniz rule. A *derivation at p* is an $\mathbb{R}$ -linear map $D : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ satisfying

$$D([f]_p[g]_p) = D([f]_p)g(p) + f(p)D([g]_p).$$

For example, $D([f]_p) = \sum_{i=1}^n v^i \partial_i f(p)$ is a derivation for each $v \in \mathbb{R}^n$: it is well defined on germs, and the ordinary product rule gives the displayed identity. Evaluation preserves products, whereas a derivation differentiates them by this rule.